Experiment 1
Electric Fields

SAFETY

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances,** attempt to repair or dismantle any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

Outline of Laboratory Exercise

Before beginning the lab, ensure you have read the Introduction to labs, appendices and other resources. This experiment will require you to use another simulation program to model the electric field and potential of a variety of configurations. The main part of this experiment will be to map the electric potential fields of various charge distributions. In today's lab you will be observing the structure and forces associated with Electric Fields. For our on campus labs, you will be exploring a range of charge distributions and observing their effects using a voltmeter and mapping your values on a grid.

For a quick review on electrical fields visit this site: http://hyperphysics.phy-astr.gsu.edu/hbase

NOTE: The terms **voltage**, **potential** and **potential difference** are all words related to the same concept. **Electric potential** is short for 'electrical potential energy per unit charge'. But be careful – it is easy to confuse potential ('potential energy per unit charge') with potential energy; they are **not** the same thing.

Potential difference is the difference in electric potential between two points in an electrical circuit (or, indeed, in a material or even in free space). **Voltage** is a term commonly used when we talk about electrical circuits. It is the same as potential difference.

Be careful that you understand what is going on.

✓ ***Pre-lab exercises****: Read the laboratory exercise and investigate the Colorado University PhET applet (instructions on the next page), then complete the online prelab questions.*

Learning Goals

- To map a 2 dimensional field.
- To competently use electrical equipment such as power supplies and voltmeters.
- To use a series of careful observations to develop an understanding of a physical phenomenon.
- To develop an understanding of electrical charges and their behaviour.

✓ **Before attempting the online Pre-lab questions**

Access the Colorado University **PhET** applet, which you should find at: https://phet.colorado.edu/en/simulation/charges-and-fields_

Play around with the app. Place a positive (red) and a negative (blue) charge in the middle of the screen. You can see the electric field lines and strength by placing a yellow sensor anywhere on the screen. The red arrow shows the size and direction of the field. On the left you can also pick up a voltmeter to measure the electric potential and a tape measure to measure the distance. Clear your screen using the orange button at the bottom of your screen.

Clear your screen, turn off the Electric Field and add **two positive charges**. For at least two different locations of the 'test charge' (the yellow sensor), investigate the direction of the force vector. Turn the Electric Field on to view the direction of the electric field lines.

You can turn the Electric Field off again and clear the screen. Now add **two negative charges**. For at least two different locations of the 'test charge' (the yellow sensor), investigate the direction of the force vector. Turn the Electric Field on to view the direction of the electric field lines.

Introduction

Most of what we know about electricity has been discovered only over the past two hundred years. But there is one aspect of electricity that people have known for much longer: static electricity. For centuries, people have been annoyed by the zap of static electricity. You may have felt a shock when touching metal after walking across a carpet; you may have noticed your clothes stuck together after they were tumbled in a clothes dryer. These are both common examples of static electricity.

Like any other kind of electricity, static electricity arises from the interaction of electric charges. There are two different kinds ('signs') of electric charge, positive and negative. Charges with the same sign repel each other, while charges with opposing signs are attracted to each other. Whether repulsive or attractive, the magnitude of the force that each charge experiences is given by:

$$F = k\frac{q_1 q_2}{r^2}$$

Here q_1 and q_2 are the two charges, r is the distance between the two charges, and the constant k = 9.0 x 10^9 N m^2 C^{-2}. The direction of the force is along a straight line, r, between the two charges.

As well as this electric force, we also talk about the electric field produced by a charge. The electric field E at a point in space is the force exerted on a positive charge at that point, divided by that charge. This is a way of describing all the possible forces that could be exerted by a single charge – the field is the force 'per charge':

$$E = \frac{F}{q} = \frac{k\frac{q_1 q_2}{r^2}}{q_2} = k\frac{q_1}{r^2}$$

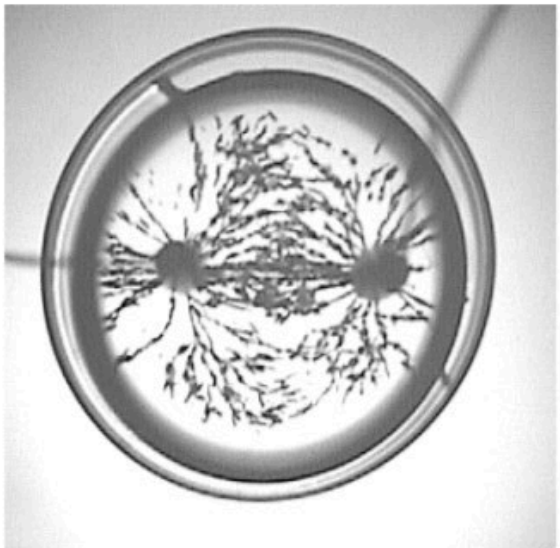
Here q_1 is the charge producing the field E, while q_2 is the charge that experiences ('feels') that field. This q_2 always cancels out (as in the equation above), because any particular field will only depend on the charge that is 'creating' it, not on anything else.

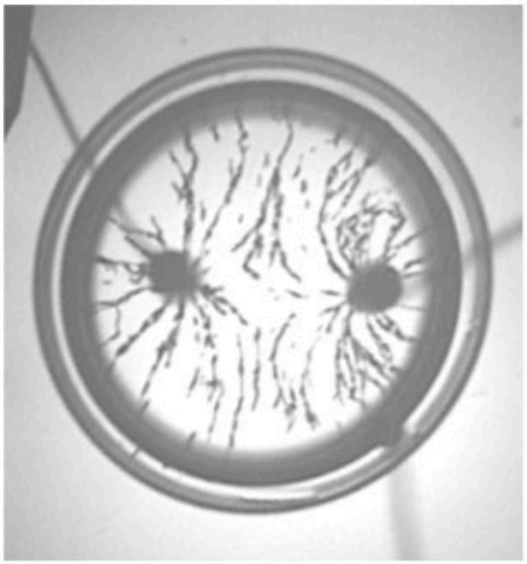
Where two charges are close together, their two fields will overlap. The total field at a particular point in space will be the two individual fields at that point, added together. (Similarly, a charge placed at this particular point will feel a total force, which is the two electric forces from the two charges added together.)

To visualise the field of a charge, we can draw a series of lines and arrows indicating the direction and magnitude of the field at various points. By convention:

- The arrows on these field lines trace the direction that a positive charge would travel in the field: away from positive charges and towards negative charges. Because $F = qE$, the direction of the field is the same as the direction of the force (on a positive charge, that is – it's the opposite direction for negative charges).
- Because of this, every single field line begins on a positive charge and ends on a negative charge.
- Field lines are drawn so that in areas where the electric field is stronger (i.e. closer to a charge), the field lines are closer together.

Field lines are a kind of metaphor – the lines themselves do not 'actually' exist, but they are a good way for us to imagine the behaviour of the field (which does exist). You may have seen field lines demonstrated in your lectures using caraway seeds in electrically-charged oil:





Experiment

In this experiment, you will investigate the electric field lines and equipotential lines in a simple 2-dimentional conducting medium. This medium consists of carbon-impregnated paper, which allows an electric current to flow. The current enters and leaves the conducting paper through metal electrodes of special geometric shapes. The shapes are: two parallel bars (straight electrodes), two discs (circular electrodes) and two points (bolt/wingnut).

You will construct two configurations with these shapes and map the electric potential field of each configuration.

Safety Tip - When wiring and re-wiring your circuit, turn off all electronic devices that are, or may be connected to the circuit.

Configuration 1:

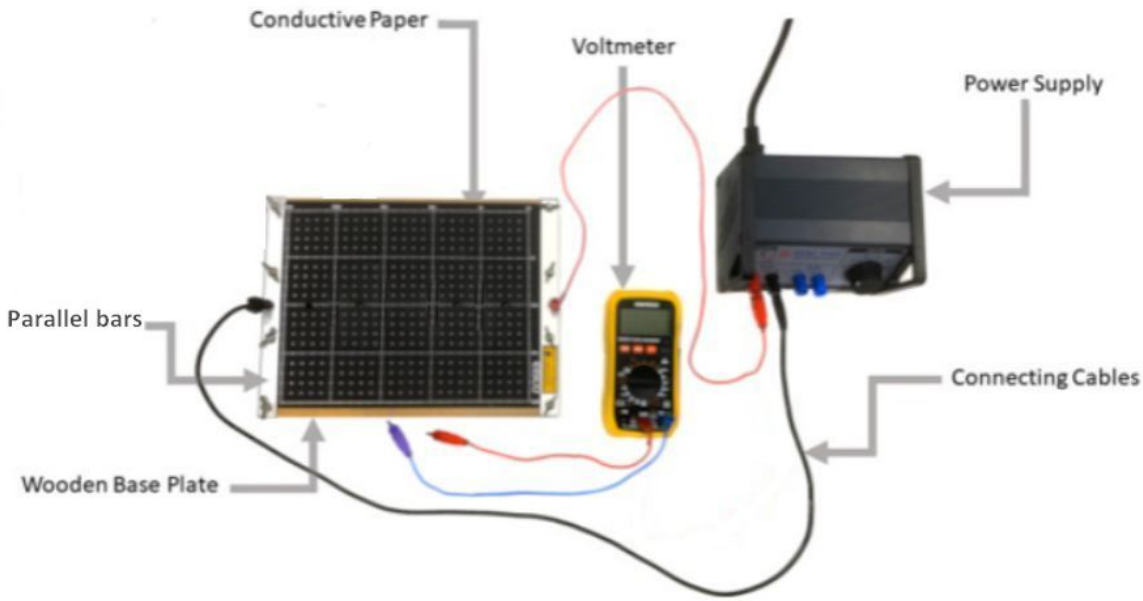
Everyone *must* complete this first configuration.

Two parallel conducting plates

The first distribution will consist of two parallel bars (or plates). This can be thought of a 2D model of a parallel plate capacitor. First remove the two discs from the wooden base plate, seen in the figure below. In this configuration, one of the two plates will be electrically connected to the positive terminal of the power supply and held at a constant positive electric potential. The other plate will be electrically connected to the negative terminal, or the "ground", of the power supply. All electric potentials are measured with respect to this electric potential (i.e. Vground = 0 V). The ground terminal of the power supply will also be connected to the ground, or COM port of the voltmeter. The probe used to measure the electric potential is connected to the "V" port of the voltmeter.

Procedure

Set up the circuit below:



Note: all wingnuts/bolts (on ANY electrodes in use, straight/circular/etc.,) need to be tight to ensure adequate contact between electrode and conductive paper.

Using the grid paper provided (which is a negative image of the conductive paper on the base plate) neatly map out the voltages in the areas between the plates.

Before you go ahead though, consider the following in your group and discuss with your demonstrator;

- Which area will you map?
- How dense should your measurements be?
- Will there be some key measurements you can make so you can quickly compare theory with experiment?
- What values to you expect to be at each of the electrodes?
- You will need to draw in the electric potentials - what process should you take when taking the voltage measurements?

Once you have your map of equipotentials, in a different colour sketch the Electric Field lines.

Configuration 2:

For the next configuration, you can select any two conductive shapes and map the resulting fields between them. You can also choose whether they have the same polarity (positive by connecting both to the positive terminal of the power supply) or opposite (one positive and one ground). The shapes can be different or the same.

Before connecting your circuit, sketch a prediction of your field map in your notes. In your sketch include a diagram of your experimental setup.

Repeat the above mapping exercise for your new configuration. If you have time, you may attempt a third configuration.

Analysis - below are some questions you *should* consider when analysing your results;

- Was the electric and potential field you mapped what you expected? Explain any differences and what might be the cause.
- Explain the structure of the fields you have created and how Coulomb's law predicts this.
- The parallel plates were constructed by two metal plates. Is this a reasonable equivalent to a capacitor? Why, why not? Can you provide possible ways to make the plates behave even more like a capacitor?
- Discuss any interesting points you found with the shapes you used.
- Are there any ways the experiment can be improved?

Only one map per configuration needs to be submitted per group, but you should sketch your map in your book and highlight key points if you are not the one submitting the full maps.

Conclusion - include a summary of your main results.